

Module 8 — Lunar Transfers & Artemis (Orbit Academy v5.5)

Printable edition. The interactive version — live simulator, self-checking questions — is the course website; open the matching tut/worksheet pages there. Dynamic diagrams appear here as bracketed figure notes.

What you will learn

- what **delta-v (Δv)** physically is, why it is the **finite currency** of spaceflight, and how to keep a **Δv logbook** for a mission.
- hands-on **maneuver intuition**: prograde, retrograde and radial burns — including the counter-intuitive rule that to **catch up with something ahead of you, you burn REVERSE**, because a lower orbit has a **shorter period**.
- how to **plan a two-burn transfer** from LEO to GEO (raise apogee, then circularize) and compute each burn's Δv from the speeds involved.
- a real respect for **timing and precision** — you will fly the same burns by hand and then watch the flight computer do it, and compare.
- why every trajectory here is drawn in the **Earth Centered Inertial (ECI)** frame — and what a co-rotating frame would show instead, and why that is sometimes useful but usually misleading.
- how **rendezvous and proximity operations** actually work: phasing along the belt, standing off, and station-keeping.
- how a **lunar transfer** differs: the target **moves**, so you lead it and aim **past** it — and it takes **three** burns, the last one cued at an apsis.

Quick-start tutorial

This is the capstone. You stop *studying* orbits and **fly** them. Two missions: first **service a satellite in geostationary orbit**, then the graduation flight — **deliver a spacecraft into orbit around the Moon**. Launch is automated; you begin already parked in a 400 km low-Earth orbit. Everything after that is yours.

The three modes — know which one you are in

The cockpit runs in three distinct modes, and mixing them up is the most common early confusion:

TRAIN — free flight. Unlimited fuel, no schedule, no score, nothing to fail. You start in LEO and just *fly*: hold thrusters and watch what each one does to your orbit. CAPCOM coaches you through drills. **Start here**. Every hour spent in TRAIN pays for itself twice in the scored missions.

PLAN — the mission planner. A full-screen overlay where you dial in each burn's Δv with the $\blacktriangle/\blacktriangledown$ steppers, watch a predicted trajectory redraw live, and check the total against your fuel budget. Nothing is flown here; nothing is spent. Iterate until the verdict says your plan works, then press $\square$ **load into cockpit & fly**.

$\square$ **FLY** — the scored mission. The clock runs, the burn cues arm, fuel actually depletes, and the flight ends in a graded debrief. This is where the plan meets your thumbs.

Picture this: TRAIN is the empty parking lot. PLAN is the map on the kitchen table. FLY is the road test — with an examiner and a finite tank of gas.

What delta-v actually is

Every maneuver in space is a **change in velocity**, and we measure the size of that change in metres per second: **delta-v**, written Δv . Fire your thruster for a while and you have changed your velocity by some Δv — that is the *whole* physical effect of the burn.

Why is it the currency? Because Δv is what your propellant buys, and the exchange rate is brutal. A rocket pushes itself forward by throwing mass backward, so the Δv you get depends on how much of your spacecraft you are willing to throw away. The relationship (Tsiolkovsky's rocket equation) is **exponential**: each additional km/s of Δv costs a larger and larger fraction of your launch mass. Doubling your Δv budget does not double the fuel — it can multiply it several times over.

The practical consequence, and the thing this module drills into you: **your spacecraft launches with a fixed Δv budget, and when it is gone, the mission ends wherever you happen to be.** There is no coasting to a gas station. A wasteful burn is not just inefficient — it is capability you no longer have.

So Δv has a physical meaning (how much you changed your velocity) *and* an economic one (how much of your finite tank you spent). A good pilot thinks in both at once. In this module your tank is **6,500 m/s** for the GEO servicing run and **9,500 m/s** for the lunar flight.

The controls — and the counter-intuitive heart of the whole module

Four thrusters, press and hold (or keys **A** / **D** / **W** / **S**):

FORWARD (prograde) speeds you up along your velocity · **REVERSE** (retrograde) slows you down · **UPWARD** pushes radially outward, away from the body you are orbiting · **DOWNWARD** pushes radially inward. □ **FINE ×0.1** (or hold **Shift**) cuts the thrust to a tenth — essential for the delicate work at the end. The , and . keys slow down and speed up **time warp**, and **F** switches the NAV reference frame.

Now the fact that trips up every newcomer. Suppose you are in a circular orbit and there is a satellite **ahead** of you in the same orbit. Your instinct says: point at it and burn FORWARD. Do that and **you fall further behind**.

Here is why. Burning FORWARD adds energy: the far side of your orbit rises, your semi-major axis a grows, and since the orbital period is $T = 2\pi\sqrt{a^3/\mu}$, your period gets **longer**. You now take more time per lap than the satellite does, so it pulls away from you.

Burning **REVERSE** does the opposite. It removes energy: your current position becomes the high point of a new, slightly smaller ellipse, the **opposite side drops**, a shrinks, and your period gets **shorter**. Now you complete each lap sooner than the satellite does, and you creep up on it — degree by degree, orbit after orbit. When you have closed the gap, you burn FORWARD to restore your period and stop drifting.

Picture this: Two runners on concentric circular tracks. To get ahead, you do not sprint — you move to the **inside lane**, where the lap is shorter. You reach the finish of each lap sooner even though you are not running faster.

One honest subtlety, because it is where people tie themselves in knots: right at the moment you fire REVERSE, your speed **drops**. It is not your instantaneous speed that wins the race, it is your **period**. (Your *average* speed around the smaller orbit, $\sqrt{\mu/a}$, does go up — but the useful way to think about it is always the period.) **Burn backwards to go forwards, because a lower orbit has a shorter lap.**

Useful numbers for later: a change δa in semi-major axis changes the period by $\delta T/T = 1.5 \cdot \delta a/a$. At GEO ($a = 42,164$ km), dropping just **400 km** shortens your period by about 1.4% — which walks you forward along the belt at roughly **5° per day**. And one degree along the GEO belt is about **736 km**, so small numbers here move you a long way.

The frame you are flying in: Earth Centered Inertial (ECI)

A trajectory has no single “true” picture — its shape depends on the frame you draw it in. For the GEO mission the cockpit uses **one** frame throughout, and the NAV panel names it: **Earth Centered Inertial (ECI)**. It does not rotate, so orbits appear in their true shapes — your parking orbit is a circle, your transfer is an ellipse with Earth at one focus, and your final orbit is a circle again. Every distance and angle you read off it means what you think it means.

It is worth knowing what the alternative would look like, because you can switch to it with **F**. A **belt-fixed** frame co-rotates with GEO — once per **sidereal day** (86,164 s), exactly the orbital period at 42,164 km. In that frame the belt and your target satellite **hold still**, and a correctly flown geostationary orbit simply *parks*: it is the only picture in which the word “geostationary” is visible at all. Take a look once, at the end, when you are on station — it makes the concept click.

But it is a poor frame to fly the transfer in, and this is worth understanding: it spins a full 360° per day, so your 5¼-hour transfer gets curled through about 80° and your 92-minute parking lap is skewed ~23°. A perfectly ordinary ellipse comes out looking like a bizarre spiral. That is the frame talking, not the physics — which is exactly why we fly in ECI and only visit the co-rotating view to make a point.

The lunar mission is drawn in the **Earth-Moon co-rotating** frame, which turns once per lunar month so the **Moon holds still**. There the rotation rate is slow compared with your flight, so it distorts little and it makes the encounter geometry far easier to read.

Mission 1 — the two-burn transfer to GEO

Moving to a much higher orbit is always a **two-step** maneuver. First, **burn prograde** in LEO to stretch your orbit until its far side (apogee) reaches GEO altitude — you have raised the apogee but you are on a long ellipse, not a circle. Then, when you coast up to that apogee, you are moving far **too slowly** to hold a circular orbit there, so you **burn prograde again** to speed up and **circularize**.

That in-between ellipse has a name you will hear constantly: the **geostationary transfer orbit — GTO**. Perigee down at LEO altitude, apogee touching the GEO belt. Real launch vehicles usually release communications satellites *into GTO* and leave the circularization burn to the satellite’s own engine — so real GEO satellites fly exactly the mission you are about to fly, minus the pilot.

The numbers you will confirm in the planner: in LEO you circle at **7.67 km/s**. Raising apogee to GEO means burning up to **10.07 km/s** — a Δv of about **2,400 m/s**. You then coast outward, slowing to just **1.62 km/s** at apogee. GEO's circular speed is **3.07 km/s**, so circularizing costs another **1,460 m/s**. Total $\approx$ **3,860 m/s**.

But reaching the belt is not the mission. **The mission is to rendezvous with a specific satellite**, and you arrive about **8° behind it** — some 5,900 km away along the belt. Closing that gap is the third phase of the flight, and it is pure counter-intuitive orbital mechanics: dip a few hundred km below the belt, let your shorter period walk you forward, then climb back and stop. That is what the rest of your fuel is for.

Keep a Δv logbook (fill this in as you plan and fly)

[Figure: Your Δv logbook. Rows 1-2 come from the planner before you fly; rows 3-4 you fill in from the fuel gauge as you actually fly them. The planner only covers getting to the belt — the rendezvous is charged to the same tank.]

Mission 2 — Fly me to the Moon (unlocks after a successful GEO run)

The graduation flight, and it is harder for two reasons. First, **the target moves**: the Moon covers about 13° of its orbit per day and your coast takes roughly 4 days, so you must depart when the Moon is about **120° of lead** ahead of where you will cross its path — like a hunter firing ahead of a flying bird. Second, you must aim your apogee just **past** the Moon (~395,000 km, beyond its 384,400 km distance); a dead-centre aim is a collision course.

The burns change character too. **Trans-Lunar Injection (TLI)** is a big prograde burn (~**3,090 m/s**) out of LEO — familiar, just larger. But the arrival burn, **Lunar-Orbit Insertion (LOI)**, points in **REVERSE**: you sweep past the Moon far too fast to be held, so you **brake** near closest approach and let lunar gravity capture you. Brake too little and the Moon slings you onward; brake too much and you drop onto the surface.

And this mission genuinely takes **three** burns. Capture is not a circular orbit — it is a captured *ellipse*. So once you are bound to the Moon the cockpit cues **burn 3: CIRCULARIZE**, timed for your next **apsis** (your high or low point). That timing is not arbitrary: at an apsis your velocity is already perpendicular to your radius, so a single along-track burn rounds the orbit out. Anywhere else, part of your motion is radial and “make it circular here” becomes a chase that never quite converges.

TLI is a **knife edge**: change it by ± 1.5 m/s and you can turn a clean capture into a lunar impact or a miss. Do not be discouraged — that sensitivity is the lesson, and it is why real missions carry course-correction burns and a flight computer.

How you are scored, and how you can fail

Reaching the target box is not enough — you must **hold** it for one full orbit, so there is room to trim and no way to fluke it.

GEO success requires all three at once, held for a full orbit: within **5%** of GEO radius, within **4%** of circular speed, and within **3.5°** of the target satellite. **Grade** = 40% altitude accuracy + 30% circularity + 30% angular closeness to the target.

Lunar success requires being gravitationally **bound** to the Moon, inside its sphere of influence (~66,000 km), within **8%** of local circular speed, held for a full revolution — then you get a victory lap. **Grade** = 50% closeness to the 3,000 km target altitude + 50% circularity.

Grades run **A** ≥ 90 , **B** ≥ 75 , **C** ≥ 60 , **D** ≥ 40 , else **F**. Ways to fail outright: re-entering the atmosphere, hitting the Moon, escaping the system, running the tank dry while off-target — or closing within **100 m** of the target satellite, which is a collision, not a rendezvous. Servicing means station-keeping *alongside*, typically a degree or two off, not contact.

What this model does and does not include

Honest limits, so you do not over-generalise: the simulator is **planar (2-D)** — every orbit shares one plane, so there are no plane-change burns or inclination costs, which in reality are expensive. Earth and Moon are **point masses** on a circular mutual orbit; there is no atmospheric drag, no Earth oblateness (J2), no solar gravity or radiation pressure. Real GEO satellites need regular station-keeping precisely because of the effects we have left out. What is faithful is the part that matters here: energy, period, timing, and the price of every maneuver in Δv .

Cockpit reference — controls and instruments

Everything you need to fly. Keep this open the first few flights.

Control / instrument	What it means	How you use it
FORWARD / REVERSE keys A / D	Thrust along (+) or against (–) your velocity. Changes your orbit’s size and period.	The workhorses. FORWARD raises the far side and <i>lengthens</i> your period; REVERSE drops the far side and <i>shortens</i> it.
UPWARD / DOWNWARD keys W / S	Thrust radially out from / in toward the body you are orbiting.	Mostly rotates and reshapes the orbit rather than resizing it. Useful for the radial component of a circularizing burn, and for sidestepping.
▢ FINE ×0.1 or hold Shift	One tenth of the thrust.	Indispensable for the last stage of a rendezvous, where the whole burn you want may be only a few m/s.
BURN NOW / ▢ BURN IN	The scheduled-burn annunciator. Amber counts you down; red is the order.	It spells out the Δv and the direction. Miss the window and it re-cues you one orbit later.

Control / instrument	What it means	How you use it
THIS BURN Δv meter	The net velocity change of the burn you are flying, signed against the ordered direction, with a marker at the target.	Pulse to sneak up on the marker. Overshoot and a REVERSE tap walks the number back <i>down</i> — it is a net, not a tally. It holds ~5 s after your last input, so tap-read-tap works, then clears.
▢ Δv BUDGET gauge	Fuel remaining, in m/s of Δv . The white tick is where the plan says you should be.	Strictly additive — a correction costs propellant exactly like the overshoot did, so this only ever falls. The gap between it and the meter is the price of sloppy flying.
© RENDEZVOUS DIRECTOR	On final approach: range, closing rate, degrees along the belt, altitude error, drift rate — plus an instruction.	Fly what it says. It is the fastest way to internalise the lower-orbit-catches-up rule.
NAV panel + frame label	Your trajectory in the named reference frame — normally Earth Centered Inertial (ECI) ; planned path (amber) vs. actual (green).	ECI shows true orbit shapes, so read distances and angles straight off it. F switches to the belt-fixed view, worth one look on station to see “geostationary”.
■ GATES ON/OFF	Wireframe hoops strung along the belt a few degrees short of the target — an ILS-style approach corridor.	Fly through the middle of each one: they turn green when threaded and red if you pass off belt altitude. Purely a piloting aid — nothing is scored on them — so switch them off with ■ GATES if they clutter the view.
AUTO FLY	The flight computer times and executes the burns for you.	Fly a mission by hand first, then watch AUTO fly the same one and compare Δv and grade.
, and . keys	Time warp down / up.	A 4-day lunar coast is unwatchable at 1×. Warp through the quiet parts; slow down before every burn.

PART A · The cockpit, the three modes, and Δv

Get oriented, understand the currency, and go fly with nothing at stake.

Exercise 1 — Find your way around: TRAIN, PLAN, FLY

Open the simulator. Three buttons at the top right decide what kind of session you are in: **TRAIN** (free flight, unlimited fuel, nothing scored), **PLAN** (the mission planner — nothing is flown or spent), and **FLY** (the scored mission, with a running clock and a real fuel gauge).

Two habits to build immediately. First, **the time-warp keys**: **⏸**, slows time down and **⏩**, speeds it up. Orbital mechanics happens on scales from seconds to days — a GEO transfer takes about 5¼ hours and a lunar coast about 4 days — so you will warp through the quiet stretches and slow right down before every burn. Second, **read the NAV frame label** before you interpret any picture of your orbit.

Predict first: Before you touch anything: if a burn only lasts a few seconds but the coast between burns lasts four days, what does that tell you about how a mission like this must be presented to a pilot?

Try it in the simulator:

Open the tutorial. Press **TRAIN**, then tap **⏸** a few times and watch the clock and the NAV map speed up; tap **⏩**, to bring it back. Find these panels and name them out loud: the window, the NAV map (note its frame label), the Δv budget gauge, the THIS BURN Δv meter, and the thrust pad. Now press **PLAN** to see the planner, and **FLY** to come back.

What to look for: the three modes are separate worlds: TRAIN cannot be failed, PLAN cannot spend fuel, FLY is scored. Time warp compresses coasts so a four-day flight is watchable, and every instrument is labelled with what frame or reference it uses.

While you watch, think about:

- Why is it valuable to have a mode where fuel is unlimited and nothing is scored?
- What could go wrong if you were still at high time warp when a burn cue arrived?

Check your understanding: What is the difference between PLAN mode and FLY mode?

- PLAN flies the mission slowly, and FLY runs the same thing at full speed
- PLAN designs and tests burns for free; FLY executes them with fuel and a clock
- They are the same mode with a different colour scheme on the panel
- PLAN is for the lunar mission, and FLY is for the GEO servicing mission

Exercise 2 — Δv — what it is, and why the tank is the mission

A burn does exactly one thing: it **changes your velocity**. We measure how much by the size of that change in m/s — the **delta-v**. That is the physical meaning: Δv is the velocity change your engine delivered.

It is also the currency, because propellant is what buys it, at an exponential exchange rate. A rocket accelerates by throwing mass overboard, so every extra km/s of Δv costs a steeply larger fraction of your launch mass (Tsiolkovsky's rocket equation). This is why

spacecraft are designed around a **Δv budget** and why mission planners argue over tens of m/s.

In this simulator your tank is stated in Δv directly: **6,500 m/s** for the GEO run, **9,500 m/s** for the lunar flight. Spend it and the mission ends where you are — the $\square$ gauge is your remaining capability, not just a number.

Predict first: Which do you think costs more Δv : (a) raising your orbit's far side from LEO all the way up to GEO, or (b) circularizing once you arrive there? Or are they about the same?

Try it in the simulator:

In **PLAN**, look at the **total Δv** readout and the budget beside it. Dial **Δv_1** up and down with the **$\blacktriangle/\blacktriangledown$** steppers and watch the total and the predicted path change. Deliberately dial in an absurd value and watch the plan be rejected as unaffordable or off-target.

What to look for: the plan is checked against a fixed budget before anything is flown. Δv is a hard constraint, not a suggestion — and the two burns of a GEO transfer are surprisingly comparable in cost.

While you watch, think about:

- Why would doubling a spacecraft's Δv budget cost far more than twice the propellant?
- Your tank is 6,500 m/s and the plan totals about 3,860. Why is that margin not "wasted" fuel?

Check your understanding: What does "delta-v" physically measure, and why is it treated as a currency?

- The spacecraft's top speed; a currency because going faster always costs more
- The change in velocity a burn produces; propellant buys it at an exponential rate
- The distance covered during a burn; a currency because distance consumes fuel
- The thrust the engine develops; a currency because thrust is what you pay for

Exercise 3 — Free flight: get your hands on the thrusters

Before any scored mission, spend real time in **TRAIN**. Unlimited fuel, no schedule, no grade. You cannot break anything — if you dip into the atmosphere you simply respawn in LEO.

The four thrusters: **FORWARD** and **REVERSE** along your velocity (keys **A / D**), **UPWARD** and **DOWNWARD** radially out and in (keys **W / S**). Press and hold. Watch the **orbit now** readout — your live perigee, apogee and eccentricity — while you thrust. That readout is the fastest teacher in the cockpit.

Predict first: Hold **FORWARD** for a couple of seconds in a circular orbit. Which part of your orbit do you expect to change: where you are now, or the opposite side?

Try it in the simulator:

Press **TRAIN**. Hold **FORWARD** for ~2 s and watch the live peri/apo numbers. Then **REVERSE** the same amount. Then try **UPWARD** and **DOWNWARD**. Use **,** and **.** to speed through a lap and see the shape you made. Do this until you can predict the readout before you release the button.

What to look for: an along-track burn barely moves the altitude where you are, but strongly changes the **opposite** side of the orbit — and therefore the size, period and eccentricity. Radial burns behave quite differently, rotating and reshaping rather than resizing.

While you watch, think about:

- Why does a prograde burn change the far side of your orbit rather than where you are?
- What single readout tells you the most about what a burn just did?

Check your understanding: You hold FORWARD briefly in a circular orbit. What happens?

- You climb immediately at your current position, gaining altitude right away
- The far side of the orbit rises — you are on an ellipse with apogee half a lap ahead
- Nothing changes at all until you make a second, matching burn later on
- Your orbit shrinks, because the added speed pulls you closer to the Earth

PART B • Maneuver intuition (in TRAIN mode)

Feel what each thruster does — including the backwards-to-go-forwards rule.

Exercise 4 — Burn low, gain high — the Oberth effect

Where you burn matters as much as how much. A prograde burn made at **perigee** — your lowest, fastest point — raises your apogee far more than the same Δv spent elsewhere. This is the **Oberth effect**: the faster you are already moving, the more orbital energy a given Δv adds.

This is exactly why the LEO→GEO transfer starts with a burn *at* the parking orbit and not somewhere convenient, and why the cockpit cues each burn at a specific point rather than “whenever you like”.

Predict first: You have a fixed 100 m/s to spend. Will your apogee end up higher if you spend it at perigee, or spread out around the orbit?

Try it in the simulator:

In **TRAIN**, make yourself a mildly elliptical orbit with a short FORWARD burn. Now coast to **perigee** (watch the live readout; CAPCOM will tell you when you are there) and hold FORWARD for 2 s — note the new apogee. Reset by burning REVERSE at perigee. Now spend a similar 2 s of FORWARD near **apogee** instead and compare how much the orbit grew.

What to look for: the same thruster hold buys much more apogee when spent at perigee. Burn placement is a lever, not a detail.

While you watch, think about:

- Why does the same Δv add more orbital energy when you are already moving fast?
- How does this shape the design of a real transfer, where the burn points are chosen first?

Check your understanding: Why is a prograde burn most effective at perigee?

- Gravity is weaker at perigee, so the burn meets less resistance there
- You are moving fastest there, so a given Δv adds the most energy — the Oberth effect
- The engine develops more thrust in the denser environment at low altitude
- It is not more effective; where you burn makes no difference to the outcome

Exercise 5 — The counter-intuitive one: to catch up, drop down

This is the single most important idea in the module, and it feels wrong the first time. You are in a circular orbit; a satellite sits **ahead** of you in the same orbit. You cannot simply aim at it and accelerate.

Burn **FORWARD** and you add energy: the far side rises, your semi-major axis a grows, and because $T = 2\pi\sqrt{a^3/\mu}$, your **period gets longer**. Each lap now takes you more time than it takes the target, so it pulls **away**.

Burn **REVERSE** and the opposite happens: the far side drops, a shrinks, your **period gets shorter**, and you come around sooner than the target does — closing the gap a little every orbit. When you have caught up, burn **FORWARD** to restore your period and stop the drift. That is a **phasing maneuver**, and it is how satellites are genuinely walked to new slots.

Be precise about the mechanism, because a half-remembered version will mislead you: at the instant you fire REVERSE your **speed drops**. What gains you ground is the shorter **period**, not a higher speed. (The average speed $\sqrt{\mu/a}$ on the smaller orbit is indeed higher — but think in periods.) The quantitative handle: $\delta T/T = 1.5 \cdot \delta a/a$. At GEO, going 400 km low shortens your period $\sim 1.4\%$ and walks you forward about **5° per day**, where $1^\circ \approx 736 \text{ km}$ of belt.

Predict first: You are 8° behind a target in the same circular orbit and you want to close the gap. Which thruster do you hold, and what will happen to your period?

Try it in the simulator:

In **TRAIN**, note your position against a landmark on the NAV map and your period from the live readout. Hold **REVERSE** briefly and watch the apogee/period **drop**. Warp forward with **.** for a few laps and watch yourself creep **ahead** of where you would have been. Then hold **FORWARD** by the same amount to restore the period and stop drifting.

What to look for: REVERSE shortens the period and walks you forward relative to anything in the original orbit; FORWARD lengthens it and lets things ahead pull away. Stopping the drift is a second, equal-and-opposite burn — a phasing maneuver is always two burns.

While you watch, think about:

- Explain to someone else, in one sentence, why burning backwards moves you forwards.
- If dipping 400 km gives $\sim 5^\circ/\text{day}$, roughly how long to close an 8° gap? What would going 800 km low do — and what does that cost you in Δv and in risk?
- Why must a phasing maneuver always end with a second burn?

Check your understanding: You are behind a satellite in the same circular orbit and want to catch it. What do you do, and why does it work?

- Burn FORWARD, since adding speed is how you close a gap ahead of you
- Burn REVERSE — the lower orbit has a shorter period, so you gain on it each lap
- Burn UPWARD, rising above the orbit so you can cut the corner across
- Nothing will work; you have to wait for the target to come round to you

Exercise 6 — Radial burns — UPWARD and DOWNWARD do something else

Along-track burns (FORWARD/REVERSE) change the orbit's **size and period**. Radial burns (UPWARD/DOWNWARD) mostly **rotate** the orbit — they move where the high and low points sit without changing the size nearly as much.

They matter for one specific job here. To make an orbit circular you must cancel *both* the excess along-track speed *and* any radial motion. At an **apsis** (your highest or lowest point) the radial motion is already zero, which is why circularizing is a single clean along-track burn there — and messy anywhere else. That is exactly why the lunar mission cues burn 3 at an apsis.

Predict first: If you burn UPWARD, do you expect your orbit to get bigger, or to rotate so that its high point moves?

Try it in the simulator:

In **TRAIN**, get roughly circular, then hold **UPWARD** for ~2 s and watch the live peri/apo: note how the high/low points shift around the orbit rather than the orbit simply growing. Try **DOWNWARD**. Then coast to an apsis and confirm that a short FORWARD/REVERSE burn there changes size cleanly with little effect on shape.

What to look for: radial burns rotate and reshape the orbit; along-track burns resize it. At an apsis the two effects separate cleanly, which is why real circularization burns are placed at apsides.

While you watch, think about:

- Why is “make my orbit circular right here” ill-posed unless you are at an apsis?
- Where would you place a burn intended purely to change your orbit's size?

Check your understanding: Why does the cockpit cue the lunar circularization burn at an apsis?

- Because the spacecraft passes closest to the Moon at that point in the orbit
- Because at an apsis your velocity is already perpendicular to the radius
- Because the thrusters run more efficiently at that part of the orbit
- It is arbitrary — any point on the orbit would serve equally well

PART C · Plan the GEO transfer

Compute both burns, log the Δv , check it against the tank.

Exercise 7 — Burn 1 — raise apogee to GEO (log the Δv)

Into the planner. GEO sits at a radius of **42,164 km** from Earth's centre — the one distance where the orbital period equals a **sidereal day** (86,164 s), so a satellite there keeps station over a fixed spot on the ground.

You are in a 400 km parking orbit, circling at **7.67 km/s**. Burn 1 is a prograde burn at that low, fast point (Oberth) that stretches your orbit until apogee just touches the belt. Vis-viva gives the speed you need at perigee for that ellipse: **10.07 km/s**. So $\Delta v_1 = 10.07 - 7.67 \approx \mathbf{2,400\ m/s}$.

Predict first: To reach 42,164 km at apogee you must speed up from 7.67 km/s. Do you expect the required Δv to be a few hundred m/s, a couple of thousand, or tens of thousands?

Try it in the simulator:

In **PLAN** with **Service a GEO satellite** selected, dial Δv_1 with the $\blacktriangle/\blacktriangledown$ steppers until the readout shows your apogee reaching GEO ($\sim 42,164$ km). Watch the gold apogee marker walk out toward the belt ring as you dial. Record Δv_1 in logbook row 1.

What to look for: about 2,400 m/s buys the climb, and the predicted path is a long ellipse — a GTO — with perigee still down at 400 km. You have not arrived anywhere yet; you have aimed.

While you watch, think about:

- Why is this burn placed in the low parking orbit rather than partway up?
- After burn 1 you are on an ellipse touching GEO. Why are you not finished?

Check your understanding: Burn 1 takes you from 7.67 to about 10.07 km/s. What has it accomplished?

- You are now in a circular orbit at GEO altitude, ready to hold station
- You are on an elliptical transfer orbit whose apogee reaches GEO altitude
- You have exceeded escape speed and left Earth's gravity behind entirely
- You have matched the target satellite's speed and can now close on it

Exercise 8 — Burn 2 — circularize at GEO (log the Δv)

Coast out to apogee and you have slowed dramatically — angular momentum is conserved, so climbing costs speed. You arrive at **1.62 km/s**. A circular orbit at GEO requires **3.07 km/s**. You are moving far too slowly to stay, and if you do nothing you fall right back to your LEO perigee.

So burn 2 is another **prograde** burn, at apogee, closing that gap: $\Delta v_2 \approx 3.07 - 1.62 = \mathbf{1,450\ m/s}$ (the planner will refine it to $\sim 1,460$ for the transfer you actually built). It surprises people that circularizing costs nearly as much as the climb did — but it is the whole reason GTO exists as a stage in real missions.

Predict first: You arrive at apogee at 1.62 km/s and need 3.07 km/s. Before computing: does this second burn cost more or less than the first?

Try it in the simulator:

In the planner's **STEP 2**, dial Δv_2 until "your new speed" matches the circular speed shown (~ 3.07 km/s). Notice the **suggested Δv_2** line recompute every time you change Δv_1 — your arrival speed depends on the transfer you actually built. Record it in logbook row 2.

What to look for: circularizing costs $\sim 1,460$ m/s, nearly as much as the raise. Two burns of comparable size, for two completely different jobs: one to aim, one to arrive.

While you watch, think about:

- Why do you arrive at apogee moving so slowly?
- What happens if you skip burn 2 entirely?
- Real launchers often drop satellites in GTO and let them circularize themselves. Why might that division of labour make sense?

Check your understanding: Why is the circularization burn at GEO so large ($\sim 1,460$ m/s)?

- GEO has stronger gravity, so more speed is needed to stay in orbit there
- You arrive at apogee at only 1.62 km/s and need 3.07 km/s to hold circular
- The spacecraft is heavier by that stage, so the same Δv costs more fuel
- Because rocket engines are markedly less efficient at high altitude

Exercise 9 — Total the logbook — and leave room for the rendezvous

Add your two entries: $\sim 2,400 + \sim 1,460 \approx \mathbf{3,860\ m/s}$. Your tank is **6,500 m/s**, so you have roughly **2,640 m/s** of margin — and you are going to use a slice of it.

This is the point most people miss on their first flight: **the planner only gets you to the belt**. You arrive about **8° behind** the target satellite ($\sim 5,900$ km along the belt), and closing that gap is a phasing maneuver you fly by hand, from the same tank — plus trims and station-keeping at the end. The white tick on the fuel gauge marks where the plan says you should still be; everything to the right of it is your reserve.

This is the payoff of keeping a logbook: **before** you light the engine you know whether the mission is affordable, and how much you are holding back for the part that is not in the plan.

Predict first: Your plan totals $\sim 3,860$ m/s against a 6,500 m/s tank. Is that margin generous, or about right, given that you still have to close 8° along the belt?

Try it in the simulator:

Sum your logbook. In the planner, confirm "total Δv " reads about **3,860 / 6,500** m/s and the verdict says the plan is on target. Press ► **GO — run sim** to watch the predicted flight play out. If you are stuck, □ **solve it for me** loads the verified reference plan — but study *why* its numbers work before flying them.

What to look for: the two planned burns fit comfortably, and the remaining margin is not slack — it is the budget for the rendezvous, the trims and station-keeping.

While you watch, think about:

- If your plan had totalled 6,400 m/s, would it be "affordable"? What would you be unable to do?
- Why does the fuel gauge mark the planned spend as well as the total?

Check your understanding: Your logbook totals $\sim 3,860$ m/s against a 6,500 m/s budget. What does the remaining $\sim 2,640$ m/s pay for?

- Nothing in particular — it is simply unused margin held back for safety
- The phasing maneuver to close the $\sim 8^\circ$ gap, plus trims and station-keeping
- The return trip to Earth once the servicing task has been completed

- The launch from the ground, which is accounted for in the same budget

PART D • Fly the GEO mission

Frames, execution, timing, the rendezvous — and how you are graded.

Exercise 10 — Before you fly: the frame you are flying in

You fly this mission in **Earth Centered Inertial (ECI)** — a non-rotating frame centered on Earth — and the NAV panel says so. Because it does not rotate, orbits appear in their true shapes: circle, transfer ellipse, circle. Angles and distances read straight off the map.

Press **F** and you can borrow the **belt-fixed** view, which co-rotates with GEO (once per sidereal day, 86,164 s). There the belt and the target **stand still** and a correct geostationary orbit simply parks — the only picture in which “geostationary” is visible. It is a great thing to look at *once*, on station at the end.

It is a bad frame to fly the transfer in, though, and knowing why is the lesson: spinning 360° per day, it curls your 5¼-hour transfer through ~80° and skews your 92-minute parking lap ~23°, turning an ordinary ellipse into a spiral. If a path ever looks bizarre, check the frame label before you doubt your flying.

Predict first: In a frame that rotates exactly once per sidereal day, what shape does a perfect circular GEO orbit trace? What would your 92-minute LEO parking orbit look like in that same frame?

Try it in the simulator:

Load your plan and fly. Note the NAV frame label — **Earth Centered Inertial (ECI)** — and watch your transfer draw as a clean ellipse. Press **F** once mid-transfer to see that same ellipse curl up in the belt-fixed view, then **F** again to come back. Save one more look for when you are finally on station beside the target.

What to look for: ECI shows true orbit shapes, which is why you fly in it. The co-rotating view answers a different question — “am I holding station?” — and is unreadable for a transfer. Neither is more correct; the label tells you which question is being answered.

While you watch, think about:

- Why is “geostationary” invisible in an inertial frame?
- Why would drawing a 92-minute LEO orbit in the belt-fixed frame be a poor way to teach transfers?
- When you are parked next to the target, which frame would you rather have on screen, and why?

Check your understanding: You fly in ECI. If you press F to view the belt-fixed (GEO co-rotating) frame while on station, a correctly flown circular GEO orbit appears as...

- A large circle once per day
- A stationary point — you park next to the target
- A curled spiral
- An ellipse with the Earth at one focus

Exercise 11 — Fly the burns by hand

Load the plan into the cockpit and fly it. The mission runs on a time-warped clock: it fast-forwards the coasts and eases toward real time as each burn approaches. An amber **BURN IN m:ss** countdown arms first, then the red **BURN NOW** order — which spells out the Δv and the direction, e.g. **2,400 m/s FORWARD**.

Two gauges, and the difference between them matters. The **THIS BURN Δv** meter shows the **net** change for this burn: if you overshoot and tap REVERSE, it walks back *down* toward the target. The **Δv BUDGET** gauge, by contrast, only ever falls — propellant does not care which way you point, so the correction costs you just as much as the overshoot did. Watch both: the meter tells you whether the burn is right, the fuel gauge tells you what your mistakes cost. That gap is the entire economics of precise flying.

Hold that thruster and watch the **THIS BURN Δv** meter climb toward its marker; release on the number. The throttle is closed-loop — quick when you are far from the target value and gentle as you close in — so big burns are fast *and* land accurately. You can also **pulse**: tap, read, tap again. The meter keeps its running total between pulses.

Remember the warp keys. Use **.** to skip the quiet coast, and **,** to slow down before the cue so you are not trying to time a burn at 900×

Predict first: When BURN NOW flashes for burn 1, which thruster will you hold, and to what value on the meter?

Try it in the simulator:

Fly the mission. At the first cue, hold **FORWARD** until the this-burn Δv reaches your logged $\sim 2,400$ m/s, then release. Coast to apogee (warp with **.**, then slow with **,**), and at the second cue hold **FORWARD** to $\sim 1,460$ m/s. Compare what you actually spent on the fuel gauge to your logbook.

What to look for: holding the cued thruster to the cued number executes your plan, and your actual spend should match the logbook to within a little. Over- or under-burning shows up immediately as a changed orbit on the NAV map and in the live peri/apo readout.

While you watch, think about:

- What happens to your apogee if you overshoot burn 1 by 100 m/s?
- Why does the cockpit taper the throttle as you approach the target Δv ?

Check your understanding: At the first BURN NOW cue for the LEO→GEO transfer, what do you do?

- Hold REVERSE briefly, letting the orbit drop before you raise it again
- Hold FORWARD until the this-burn Δv meter reaches your planned $\sim 2,400$ m/s
- Do nothing and wait for apogee, where the transfer burn is cheaper
- Hold UPWARD, pushing outward toward the altitude you are aiming for

Exercise 12 — How delicate is this? Fly it by hand, then let the computer fly it

Now the point of the exercise. These maneuvers are **timed and sized precisely**, and a burn that is a second late or fifty m/s over lands you somewhere measurably different. That is not the simulator being fussy — it is why real missions are flown by computers with humans supervising, and why they carry course-correction burns.

So run the comparison honestly. Fly the mission by hand and note your grade and total Δv . Then fly the *same* mission with **AUTO FLY** engaged: the flight computer fires at each cue and releases on the number, and it will also fly the phasing and trims. Compare the two runs.

AUTO FLY is not there to do your homework — it is a demonstration of the standard you are trying to meet, and a way to see a clean flight when you are stuck. You can toggle it mid-flight and take the controls back at any time.

Predict first: Before you compare: do you expect the flight computer to spend more or less Δv than you did? Where do you think your biggest losses will come from — burn size, or burn timing?

Try it in the simulator:

Fly the GEO mission entirely by hand and write down your grade, your total Δv , and your final separation from the target. Then ♡ **Fly again** with **AUTO FLY** on and record the same three numbers. Finally, fly it once more by hand, watching the cues the way the computer did.

What to look for: the computer spends noticeably less and lands closer, mostly by releasing exactly on the number and burning exactly at the cued instant. Hand-flying loses Δv to overshoot and to late burns — and the losses compound, because every error must be corrected later out of the same tank.

While you watch, think about:

- Where did you lose the most: the size of your burns, or their timing?
- Why does a small error early (say in burn 1) cost more than the same error late?
- Given how tight this is, why do human pilots remain in the loop at all?

Check your understanding: What is the main lesson of flying the mission by hand and then with AUTO FLY?

- That the autopilot is effectively cheating, since it knows the answers
- That these maneuvers need precise sizing and timing — early errors compound
- That hand-flying always produces a better result than the autopilot does
- That the Δv budget does not really constrain what the mission can do

Exercise 13 — The rendezvous: phase in and hold station

You are circular at GEO — and about **8° behind** the target satellite, roughly 5,900 km along the belt. Look out the front window: the red bead with the cyan reticle and a range readout is your rendezvous. Now apply Part B.

The maneuver, in three moves. (1) Hold **REVERSE** for ~15 m/s: that drops you a few hundred km below the belt, shortening your period so you **drift forward** at about 5°/day. (2) **Coast** — warp with . and let geometry do the work; thread the ILS gates on the way in. (3) As you arrive, hold **FORWARD** ~15 m/s to restore a one-sidereal-day period and **stop the drift**, then trim with □ **FINE** until you are circular alongside.

The ☉ **RENDEZVOUS DIRECTOR** (bottom right) gives you every number you need: range, closing rate, degrees along the belt, altitude error, drift rate — and a plain-language instruction. Fly what it says and watch how it matches the physics you learned in TRAIN.

Two aids for this phase. The  **GATES** button toggles a string of wireframe hoops along the belt, a few degrees short of the target — an approach corridor borrowed from aircraft ILS. Fly through the middle of each: they turn **green** when threaded, **red** if you cross off belt altitude. They score nothing; they are there to make your approach geometry visible, and you can switch them off if they clutter the window. And once you are inside a few hundred km, turn on  **FINE ×0.1** (or just hold **Shift**): full thrust delivers ~50 m/s in the shortest tap you can make, which is far too coarse when the burn you want is 6 m/s. Fine thrust is what makes the last stage flyable at all.

Two cautions. Aim to sit a degree or two **behind** the target, not on top of it: closing within **100 m** counts as a **collision**, and there are neighbouring satellites on the belt too. And do not try to “fly at” the target — every attempt to shortcut the phasing will cost you fuel and leave you further out.

One more thing the director will teach you, and it is the same lesson as the lunar burn 3: if your orbit is slightly egg-shaped, you can only round it out **at an apsis** — your high or low point. So the director will sometimes say **WAIT — HIGH POINT IN 6h 00m, then hold FORWARD ~6 m/s**. Believe it, and warp the wait with .. Burning at any other point tilts the orbit instead of circularizing it, and the correction you need flips direction every half lap, so chasing it never converges.

Predict first: You are 8° behind and want to close it. Which thruster, roughly how long, and about how many days will the drift take at ~5°/day?

Try it in the simulator:

After circularizing, find the red target in the window. Read the director, then fly the phasing: ~15 m/s **REVERSE**, warp the drift, thread the gates, then ~15 m/s **FORWARD** plus  **FINE** trims to settle 1-2° behind the target. Hold it until CAPCOM confirms station. Log rows 3 and 4 of your logbook.

What to look for: a couple of tens of m/s and a day of patience closes thousands of km — and trying to hurry it costs more. Stopping the drift is as deliberate as starting it. This is real GEO practice: operators walk satellites to new slots exactly this way.

While you watch, think about:

- Why is the total Δv for phasing so small compared with the transfer burns?
- Why does the director sometimes tell you to WAIT rather than burn now — and what goes wrong if you burn anyway?
- What would happen if you dipped 1,000 km low instead of 400 — faster, but what is the trade?
- Why hold station *behind* the target rather than right at it?

Check your understanding: You are 8° behind the target on the GEO belt. What is the correct first move?

- Hold FORWARD to accelerate directly toward the satellite ahead of you
- Hold REVERSE ~15 m/s to drop below the belt, shortening your period
- Hold UPWARD to rise above the belt and cut across the gap from there
- Wait — orbital mechanics will close the gap without any burn at all

Exercise 14 — Why the director tells you to wait

Late in the approach the director may stop giving you a burn and instead say something like **WAIT — HIGH POINT IN 6h 00m, then hold FORWARD ~6 m/s**. That is not the simulator stalling: it is the single most useful habit in orbit trimming.

If your orbit is slightly egg-shaped, then “what burn would make me circular right here?” has an answer that **flips direction every half lap** and never reaches zero — because at your low point you are moving faster than circular and at your high point slower. Chase it and you will burn fuel forever while your orbit gets no rounder.

At an **apsis** your velocity is already perpendicular to your radius: no radial motion to cancel. One along-track burn there genuinely circularizes the orbit. So the correct move is to **coast** to the apsis — warp with . — and spend a few m/s at the right moment with □ **FINE**.

Predict first: Your orbit is slightly eccentric near GEO. If you burn to “fix” it at a random point in the orbit, what do you expect to happen over the next few laps?

Try it in the simulator:

On approach, watch the director’s **eccentricity** readout and its **next high/low point** countdown. Deliberately trim at a non-apsis point and watch the eccentricity refuse to fall (and the instruction flip direction half a lap later). Then reset by coasting to an apsis and applying the burn it names, with □ **FINE**, and watch eccentricity drop toward zero.

What to look for: trimming away from an apsis tilts the orbit rather than rounding it, and the demanded correction reverses each half lap. Applied at an apsis, a burn of a few m/s drops eccentricity straight toward zero.

While you watch, think about:

- Why does the “circularize here” requirement change sign between your low and high points?
- How is this the same rule that governs burn 3 of the lunar mission?

Check your understanding: The director says “WAIT — HIGH POINT IN 6h 00m, then hold FORWARD ~6 m/s.” Why not just burn now?

- Because the engine needs time to cool between successive burns
- Because away from an apsis your motion has a radial component to fight
- Because propellant goes further at apogee than anywhere else in the orbit
- Because the target satellite is not yet visible from your position

Exercise 15 — The debrief: how you are actually graded

Touching the target box is not success — you must **hold** it for a full orbit, which leaves room to trim and prevents flukes.

GEO success means all three at once, held for one orbit: within **5%** of GEO radius, within **4%** of circular speed, and within **3.5°** of the target satellite. Your **grade** weights 40% altitude accuracy, 30% circularity, and 30% angular closeness to the target — so a beautifully circular orbit in the wrong place still loses a third of the marks.

Grades: **A** ≥ 90 , **B** ≥ 75 , **C** ≥ 60 , **D** ≥ 40 , otherwise **F**. The debrief also reports your Δv spent against the budget.

Ways to fail outright: re-entering the atmosphere, escaping, running the tank dry while off-target, or colliding with the target (inside 100 m). A successful GEO run **unlocks the lunar mission**.

Predict first: Which do you expect to cost you more marks: being 200 km off GEO radius, or sitting 10° away from the target satellite?

Try it in the simulator:

Finish a GEO run and read the debrief carefully: the grade, the per-metric breakdown, and Δv spent versus budget. Note which metric cost you the most. Then fly it again and try to improve that one number specifically.

What to look for: the score is a weighted combination, not a pass/fail — and angular closeness to the target carries as much weight as circularity, which is why the phasing leg is a third of the mission and not an afterthought.

While you watch, think about:

- Why require holding the box for a full orbit rather than just reaching it?
- Which single improvement would raise your grade the most, and what would it cost in Δv ?

Check your understanding: For the GEO mission, what does the grade reward?

- Only how quickly you completed the mission from launch to station
- Altitude accuracy 40%, circularity 30%, and closeness to the target 30%
- Only how little propellant you managed to spend along the way
- Only that you reached GEO altitude at some point during the flight

PART E • Graduate — fly me to the Moon

A moving target, a knife-edge injection, and three burns.

Exercise 16 — Lead a moving target — plan the TLI burn

Everything so far aimed at a fixed slot. The Moon **moves**: about 13° of its orbit per day, and your coast out takes roughly **4 days**. So you must leave when the Moon is around **120°** of lead ahead of where you will cross its path — a duck hunter's lead, on a four-day scale.

Second twist: aim your apogee just **past** the Moon, about **395,000 km**, rather than at its 384,400 km distance. A dead-centre aim is a collision course; you want to cross *ahead* of it and let its gravity take hold.

And be warned about sensitivity. **TLI is a knife edge**: ± 1.5 m/s — a fraction of a percent — can flip a clean capture into an impact or a miss. Near-lunar energies are like that. This is the clearest demonstration in the course of why precision matters.

Predict first: The Moon moves $\sim 13^\circ/\text{day}$ and your coast lasts ~ 4 days. Roughly how far ahead of the Moon's current position should you aim, and how does that compare with the 120° lead?

Try it in the simulator:

In **PLAN** choose **Fly me to the Moon**. Dial Δv_1 so your apogee reaches **~395,000 km**, and set the **lead angle** slider near 120° . Press ► **GO — run sim** and watch the encounter. Now nudge Δv_1 by just **± 2 m/s** and run it again — watch the closest approach swing by thousands of km.

What to look for: a change of a couple of m/s in a 3,000 m/s burn completely changes the encounter. Lead angle and aim point are as important as the Δv itself, and all three must agree before the prediction shows capture.

While you watch, think about:

- Why does such a tiny fractional change in TLI produce such a large change at the Moon?
- Why aim past the Moon rather than at it?
- What would you add to a real mission design to cope with this sensitivity?

Check your understanding: Why do you lead the Moon by $\sim 120^\circ$ instead of aiming where it is now?

- To keep the spacecraft clear of the Moon's shadow during the approach
- Because it moves $\sim 13^\circ/\text{day}$ and the coast takes ~ 4 days, so it must arrive with you
- To cut the propellant needed for the eventual return trip back to Earth
- Because the thruster axes are defined differently once near the Moon

Exercise 17 — LOI — brake, or be slung right past

Here is what surprises pilots. After four days of coasting you reach the Moon and the second burn points **in REVERSE**. Why? Near apogee you are crawling (~ 0.2 km/s) while the Moon barrels along its orbit at ~ 1.02 km/s. It runs you down from behind, and you sweep past it far too fast to be held.

Lunar-Orbit Insertion is therefore a **brake**: you slow your speed *relative to the Moon* near closest approach until lunar gravity can keep you. Brake too little and it slings you onward; too much and you drop toward the surface.

Note carefully what "relative to the Moon" means: near the Moon your thrusters work in the **Moon's** frame, and the guidance asks you to brake to the local circular speed at your current distance. The plan calls for roughly **900 m/s**; flown against live guidance it typically settles nearer **650 m/s**.

Predict first: At GEO, burn 2 was FORWARD (speed up to stay). At the Moon it is REVERSE. What is different about arriving at a massive, moving body that flips the burn direction?

Try it in the simulator:

Dial Δv_2 to ~ 900 m/s (the planner flies it in REVERSE automatically), then press ► **GO**. Nudge Δv_1 (± 1 – 2 m/s), the lead angle and Δv_2 until the verdict reads ☐ **LUNAR CAPTURE predicted**. Log Δv_2 and your total against the 9,500 m/s budget. ☐ **solve it for me** loads the verified reference (3,087 / 900 / 120° lead) — study why it works before flying it.

What to look for: capture is a genuine three-body outcome, so the simulation itself grades the plan. Braking harder is not automatically safer — it can drop you onto the

surface.

While you watch, think about:

- Why is arriving “too fast” the default outcome without a brake?
- What is the physical difference between capture (an ellipse) and a circular orbit?
- Why does the planner rely on simulation here rather than a formula?

Check your understanding: Why does Lunar-Orbit Insertion point in REVERSE when circularizing at GEO pointed FORWARD?

- The cockpit relabels the thrusters once you are near the Moon
- You arrive too fast relative to the Moon to be captured, so you must brake
- LOI points in reverse because it also begins the journey back to Earth
- It does not — both the LOI and the GEO burn are prograde burns

Exercise 18 — Fly it: three burns to a lunar orbit

Load the plan and fly. **TLI** feels familiar — a big FORWARD hold at the cue. Then the clock warps through the four-day coast; watch the NAV auto-zoom through **departure** → **cruise** → **arrival**, with a cyan dashed line showing where your **current** orbit is headed and a marker for the predicted closest approach. Slow down with , as the encounter nears.

At the LOI cue, hold **REVERSE** to the cued value. Capture is not the end: what you have is a captured **ellipse**. So the cockpit now cues **burn 3 — CIRCULARIZE**, timed for your next **apsis**, where (as you learned in Part B) a single along-track burn can round the orbit out. Fly the cued components, or hand it to **AUTO FLY** and watch how it is done.

Lunar success: bound to the Moon, inside its sphere of influence, within **8%** of local circular speed, held for a full revolution — then you fly a victory lap. Your grade is 50% closeness to the **3,000 km** target altitude and 50% circularity. Budget 9,500 m/s; a clean flight uses well under half of it.

Predict first: During the burn-free four-day coast, will your path look like a straight shot to the Moon, or a long curve that the Moon swings around to meet?

Try it in the simulator:

Press ☐ **load into cockpit & fly**. Execute **TLI FORWARD** at the cue (~3,090 m/s). Ride the coast, watching the encounter geometry develop. At the LOI cue hold **REVERSE** to the cued value. Then fly the cued **burn 3 — CIRCULARIZE** at your apsis, and trim until CAPCOM confirms you have held a stable lunar orbit for a full revolution. Compare your spend to the logbook — then fly the whole thing again with AUTO FLY and compare grades.

What to look for: the “impossible” shot is your GEO skills re-aimed: one big prograde burn, a patient coast, a brake instead of a boost, and a circularization placed at an apsis. Three burns, each with a different job. When CAPCOM confirms capture you get a victory lap and then the debrief — you have flown an Artemis-class profile end to end.

While you watch, think about:

- Why does capture leave you on an ellipse rather than a circle?
- Why is burn 3 cued at an apsis rather than immediately after capture?

- Which of the three burns was hardest to fly by hand, and why?
- What did AUTO FLY do differently from you?

Check your understanding: You have braked and are captured by the Moon. Why is there a third burn?

- To leave lunar orbit again and start the return journey back to Earth
- Because capture leaves an ellipse; a burn at an apsis rounds it out
- To correct an error made during the trans-lunar injection burn earlier
- There is no third burn — being captured completes the mission itself

Key points

- You flew two real missions from a cockpit: a **GEO servicing rendezvous** and a **lunar orbit insertion** — planning each, executing it by hand, and comparing against the flight computer.
- **Delta-v** is the velocity change a burn delivers, and the finite currency of spaceflight: propellant buys it at an exponential rate, so a spacecraft is designed around a Δv budget. A **logbook** tells you before you fly whether a mission fits the tank — and reminds you to reserve fuel for the parts the plan does not cover.
- The **three modes**: TRAIN (free flight, unlimited fuel, no score — start here), PLAN (design burns, spend nothing), FLY (the scored mission).
- A LEO→GEO transfer is a **two-burn** maneuver — raise apogee ($\sim 2,400$ m/s), then circularize ($\sim 1,460$ m/s) — via the **GTO**. Circularizing costs nearly as much as the climb because you arrive at apogee moving only 1.62 km/s and need 3.07. Burns are placed at perigee/apogee deliberately: the **Oberth effect** means a given Δv buys the most energy where you are moving fastest.
- The counter-intuitive rule you will never forget: **to catch something ahead of you, burn REVERSE**. A lower orbit has a smaller semi-major axis and therefore a **shorter period** ($\delta T/T = 1.5 \cdot \delta a/a$), so you come around sooner and creep forward — about $5^\circ/\text{day}$ for a 400 km dip at GEO, where $1^\circ \approx 736$ km. Then burn FORWARD to stop the drift. Your speed at the burn point actually drops; it is the period that wins the race.
- Circularizing works cleanly only at an **apsis**, where your velocity is already perpendicular to your radius — which is why the lunar mission cues burn 3 there.
- **Frames matter**: you flew in **Earth Centered Inertial (ECI)**, which is non-rotating and therefore shows every orbit in its true shape. The belt-fixed frame (one sidereal day) is the only picture in which “geostationary” is visible — worth one look on station — but it curls a transfer into a spiral, so it is no frame to fly in. The lunar flight uses the Earth-Moon co-rotating frame, which holds the Moon still. Read the NAV label before you interpret any shape; **F** switches it.
- These maneuvers are **delicately timed and sized**. TLI is a ± 1.5 m/s knife edge. Hand-flying costs Δv to overshoot and late burns, and early errors compound — which is why real missions fly on computers with humans supervising, and carry correction burns.
- The lunar flight takes **three** burns: TLI ($\sim 3,090$ m/s prograde), LOI (a REVERSE brake near closest approach, ~ 650 – 900 m/s), and a **circularization at an apsis**. Lead the Moon $\sim 120^\circ$ and aim $\sim 395,000$ km — past it, not at it.
- Success is never just *touching* the target: you must **hold** the box for a full orbit. GEO is graded 40% altitude / 30% circularity / 30% proximity to the target satellite;

the Moon 50% altitude / 50% circularity. Failing means re-entry, impact, escape, a dry tank off-target, or coming within 100 m of the satellite you were sent to service.

Final quiz

Answers are checked in the online version.

Q1. What does delta-v physically measure, and why is it the currency of spaceflight?

- The spacecraft's maximum attainable speed; going faster is always better
- The change in velocity a burn produces; propellant buys it exponentially
- The distance covered during a burn, since covering distance costs fuel
- The thrust the engine develops, thrust being the expensive part to build

Q2. You are in a circular orbit, 8° BEHIND a satellite in the same orbit. To catch it you should...

- Burn FORWARD, because adding speed is how you close a gap ahead of you
- Burn REVERSE, because a lower orbit has a shorter period and you gain each lap
- Burn UPWARD, rising above the orbit to cut the corner across the gap
- Do nothing, because the gap closes on its own as the orbits diverge

Q3. Immediately after that REVERSE burn, what happens to your speed — and what actually lets you catch up?

- Speed rises immediately, and that rise is what closes the gap on the target
- Speed at the burn point drops; the shorter period of the smaller orbit closes it
- Speed is unchanged by the burn; only the altitude of the orbit matters here
- Speed drops and stays lower, so the gap actually widens rather than closing

Q4. Why is the GEO circularization burn (~1,460 m/s) nearly as large as the burn that got you there?

- GEO has stronger gravity, so more speed is needed to remain in orbit
- You arrive at apogee at only 1.62 km/s and must reach 3.07 km/s to hold circular
- The spacecraft is heavier by that point, so the same Δv costs more fuel
- Because rocket engines run markedly less efficiently at high altitude

Q5. You fly in Earth Centered Inertial (ECI). What would the belt-fixed (GEO co-rotating) frame show instead, and why is it not used for the transfer?

- The same shapes throughout; the frame is only a different colour scheme
- A correct GEO orbit becomes a stationary point, but the frame distorts the transfer
- A more accurate picture of the orbit, which is why it should always be used
- The same orbit with gravity removed from the underlying calculation

Q6. Why does the cockpit cue the lunar circularization burn (burn 3) at an apsis?

- Because the spacecraft passes closest to the Moon at that point
- Because at an apsis your velocity is perpendicular to the radius, so one burn rounds it
- Because the thrusters run more efficiently at that part of the orbit
- Because that is the only point where the Moon is visible from the cockpit

Q7. TLI can be flipped from a clean capture to an impact or a miss by a change of about...

- 500 m/s
- 1.5 m/s
- 100 km/s
- It cannot be changed once planned

Q8. What must you do to SUCCEED at the GEO mission (not merely reach GEO)?

- Touch 42,164 km at any speed, however briefly, at some point in the flight
- Be within 5% of radius, 4% of circular speed and 3.5° of the target, and hold it
- Spend less than 1,000 m/s of the budget getting there and staying there
- Arrive at the belt before the mission clock passes its stated time limit

Q9. You are 300 m from the target satellite and still closing at 10 m/s. What is the correct concern?

- Nothing — making contact with the satellite is the goal of a servicing run
- Inside 100 m counts as a collision; null the closing rate and hold station off it
- Speed up to complete the rendezvous before the geometry slips away from you
- Climb above the belt at once, then come back down more slowly onto the target

Q10. Which best captures why you fly a mission by hand and then with AUTO FLY?

- To prove that a human pilot can beat the computer at sizing the burns
- To feel how precisely the burns must be sized and timed before seeing it done right
- Because the autopilot solves the flight using a different set of physics
- To save time, since the automatic run is much quicker than flying by hand

Learn more

- Delta-v — Wikipedia — the quantity this whole module is about
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- Tsiolkovsky rocket equation — Wikipedia — why Δv costs propellant exponentially
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<https://www.nasa.gov/humans-in-space/artemis/>